

CS 201: Fundamentals of Computer Science II

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Introduction

- About me
 - Ph.D. in Computer Science, *Virginia Tech*
 - Research: AI-based Smart Systems, Machine Learning, Explainable AI
- **About you**
 - Your name
 - Your major
 - What was your favorite (or least favorite) topic in CS101?
 - C language

About This Course

- Transition from C to Java
- Write modular and reusable programs
- Learn the foundations of Object-Oriented Programming (OOP)
- Develop problem-solving skills using Java
- Prepare for Data Structures and Algorithms

This course is **NOT** about learning Java syntax.

It is about learning better ways to design and organize programs.

Why Java?

- Widely used language
- Object-oriented programming
- Large standard library
- Platform independent
- Used throughout this course for implementing data structures

Java vs C

C	Primitive Java
Function	Method
struct	class
printf	System.out.println
scanf	Scanner
Manual memory	Garbage Collector (later)

Primitive Java is the **C-like** subset of Java that includes the basic programming constructs needed to write simple programs.

First Java Program

```
public class Hello {  
    public static void main(String [] args){  
        System.out.println("Hello World!");  
    }  
}
```

- **class:** Defines a Java class. Every Java program is organized into one or more classes.
- **main():** The starting point of every Java application. Program execution begins here.
- **System.out.println():** Prints text to the console and moves the cursor to the next line.

Console Output

- `System.out.println("Hello");`
- `System.out.print("Hello");`
- `System.out.printf("Age = %d\n", age);`

Console Input

- **import:** Makes classes from other packages available for.
- **Scanner:** A built-in Java class used to read input from the keyboard.
- **new Scanner(System.in):** Creates a new Scanner object
- **nextInt():** Reads an integer entered by the user.
- **next():** Reads the next word entered by the user as a String.

```
import java.util.Scanner;

public class Hello {
    public static void main(String [] args){
        System.out.println("Hello World!");

        Scanner keyboard = new Scanner(System.in);

        System.out.println("What's your name?");

        String name = keyboard.next();

        System.out.println("Hello," + name);
    }
}
```


Primitive Types

Java Data Type	C Data Type
int	int
short	short
char	char
float	float
double	double

Example Program

```
import java.util.Scanner;

public class Hello {
    public static void main(String [] args){
        System.out.println("Hello World!");

        Scanner keyboard = new Scanner(System.in);

        System.out.println("What's your name?");

        String name = keyboard.next();

        System.out.println("Hello," + name);
    }
}
```

Key Takeaways

- Similarities and differences between Java and C
- The structure of a Java program
- Console input and output
- Primitive Java
- Basic concepts of classes, objects, and methods